

The Effects of Different Resistance Training Modalities on Muscle Strength in Community-Dwelling Older Adults: A Network Meta-Analysis

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Keywords

Review · Strength training · Mode · Elderly · Exercise

Abstract

Introduction: Resistance training is widely recognized as a valuable method to enhance physical function, reduce the risk of falls, and improve overall health, independence and well-being in older adults. The training can be performed through various modalities of which the comparative effects have not yet been thoroughly elucidated. This network meta-analysis (NMA) therefore aims to compare and rank different resistance training modalities regarding their effects on strength outcomes in older adults. **Methods:** The studies included in this NMA were identified through a comprehensive search across three health-related and biomedical databases (PubMed, Web of Science, and SPORTDiscus). We included (randomized) controlled trials that assessed the effect of resistance training modes (including: free weight training, machine-based training, bodyweight training,

elastic band training, and mixed training) in comparison to each other or to a control condition, focusing on traditional maximal strength outcomes in older individuals. **Results:** This NMA includes 102 trials involving 4,754 community-dwelling seniors with a mean age of 70 ± 4 years (range 63–87 years of age). All of the included resistance training modalities yield moderate to large positive effects on strength when compared to a “no intervention” control condition. Machine-based training {standardized mean difference (SMD) = 1.34 [95% confidence interval (CI): 1.16; 1.52]} elicits the largest effects, followed by free weight (SMD = 1.15 [95% CI: 0.28; 2.03]), elastic band (SMD = 0.93 [95% CI: 0.49; 1.37]), mixed (SMD = 0.88 [95% CI: 0.51; 1.25]), and bodyweight training (SMD = 0.71 [95% CI: 0.16; 1.26]). **Conclusion:** Machine-based training elicits the largest effects with the smallest variance. However, our findings demonstrate that various resistance-training modalities can have large positive effects on strength development in older adults. We suggest that, with adequately chosen load criteria such as training intensity, volume and duration

of the training period, the selection of the appropriate resistance training type might depend on individual preference, enjoyability, and practicability.

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Plain Language Summary

As people age, it becomes especially important to stay strong to maintain independence, avoid falls, and stay healthy. Resistance training – exercises that make your muscles work against some form of resistance – is a proven way to improve strength in older adults. There are many ways to perform resistance training, including using machines, free weights, bodyweight, or elastic bands. However, it's not yet fully clear which of these methods works best for building strength in older people. In this study, we looked at results from 102 previous studies that involved a total of more than 4,700 older adults, most of them around 70 years old. We compared the effects of different types of resistance training on muscle strength. We found that all the training types improved strength more than performing no exercise at all. Training with machines showed the biggest improvement in strength, followed by using free weights, elastic bands, a mix of methods, and bodyweight exercises. While machine training had the strongest effect overall, all the methods can be effective if the training is done with the right intensity and over a long enough time. This means that older adults can choose the type of resistance training that suits them best – what they enjoy and what is most practical for them – without missing out on the benefits.

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Introduction

Degenerative processes accelerate at an increased rate with the onset of the sixth decade in life, adversely affecting the neuromuscular system, leading to losses in muscle strength and muscle mass [1, 2]. These age-related declines are attributed to neural factors such as the numerical loss of alpha motoneurons and morphological factors, including a lower number and smaller size of muscle fibers, particularly type-II muscle fibers as well as their interaction [3]. Since muscular weakness is strongly linked to a decline in mobility and an increased risk of falling [4], exercise and particularly resistance training has proved effective in counteracting these declines. Thus, resistance training is widely recognized as a valuable means to enhance overall health and quality of life for older adults [5–7]. Consequently, Position Statements from the American College of Sports Medicine [8, 9] and the

National Strength and Conditioning Association [10] strongly recommend a regular resistance training routine comprising 2–4 weekly exercise sessions training all major muscle groups for older adults. However, resistance training can be performed through a variety of different training modalities, including free weight exercises, machine-based exercises, bodyweight exercises, and elastic band exercises [11, 12]. Unfortunately, meta-analyses examining the impacts of various resistance-training modalities on the strength of older adults are sparse. Limited pairwise meta-analytical findings suggest similar strength and muscle mass gains via free weight and machine [13] or elastic band [11] approaches. Further studies indicate comparable improvements with free weight versus bodyweight training [14, 15]. Most of these findings are, however, limited to younger populations and none of the studies [13–15] provide an overarching comparison of multiple resistance training modalities focused on older adults. Appropriately, the National Strength and Conditioning Association's Position Statement on resistance training for older adults acknowledges that exercise type and modality are training variables that should be taken into account but does not present any specific recommendations. This fact highlights the need for a sufficient analysis to guide informed decision-making, especially in the case of older populations. Unlike a traditional pairwise meta-analysis, a network meta-analysis (NMA) enables the calculation and comparison of treatment estimates from direct and indirect evidence by using a common comparator that multiple interventions compare against [16]. As a result, it allows the estimation of comparative effects for a variety of different interventions within one analysis and provides an estimated ranking of the different training modalities that can help identify the intervention with the largest estimated effect for a specific outcome [17]. Against this background, the present NMA aimed at examining and comparing the effects of different resistance training modalities (free weight exercises, machine-based exercises, bodyweight exercises, and elastic band exercises) on maximal strength adaptations in older populations. The overall results could have a considerable impact on the design and customization of resistance training programs for older adults.

Methods

Search and Screening Procedures

This network meta-analytical review was previously registered (PROSPERO: CRD42023399358) and conducted in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses for NMAs

Table 1. Search strategy

Search level	Search terms with Boolean operators
Search #1	"one repetition maximum" OR "maximal voluntary contraction" OR "muscle strength" OR "muscular strength" OR "maximal torque" OR "hypertrophy" OR "muscle thickness" OR "cross sectional area"
Search #2	#2 AND ("older" OR "elderly" OR "senior")
Search #3	#2 AND ("intervention" OR "training")
Search #4	#3 NOT (injur* OR cancer* OR stroke OR apoplex* OR parkinson* OR "multiple sclerosis")

(PRISMA-NMA) [18]. The PRISMA checklist is provided as online supplementary Table 1 (for all online suppl. material, see <https://doi.org/10.1159/000546346>). The literature search and screening process were conducted by multiple researchers and research assistants, with each step supervised by the lead author. Each stage of the screening involved at least two independently working collaborators. In the case of a conflict, the final decision was made by the lead author. Three health-related and biomedical databases (PubMed, Web of Science and SPORTDiscus) were screened from the inception of the respective journal until February of 2025. Relevant search terms were combined with Boolean operators (OR/AND) and applied to three search levels (Table 1). Duplicates were removed and the remaining studies underwent a manual screening. The remaining studies were gradually screened using (1) the titles, (2) abstracts, and (3) full texts of the potentially eligible articles. A comprehensive summary of the screening process is displayed in Figure 1. The inclusion criteria were carefully discussed among the authors and chosen based on relevant literature as well as similar published (network) meta-analyses. They were applied following the PICOS approach [population (P), intervention (I), comparators (C), main outcome (O), and study design (S)]: (P) participants were independently living, community-dwelling adults without any cognitive, neurological, orthopedic, and/or cardiac conditions that could affect physical testing or training with an age of ≥ 60 years when subtracting one standard deviation from the participant's mean age; (I) resistance training based interventions of ≥ 28 days duration or ≥ 8 training sessions were included; (C) active and/or inactive control group(s) that received a placebo treatment or did not receive any intervention served as the comparator; (O) included studies measured at least one maximal strength-related outcome such as 1RM (1-repetition maximum), maximal voluntary contraction, maximal force or maximal torque; (S) two- or multiarmed controlled intervention studies with pre- and post-testing and

a total sample size of $n \geq 10$ were included. Furthermore, only full-text articles published in English in a peer-reviewed journal were considered for inclusion. The exclusion criteria were: No adequate control conditions (i.e., integration into the network was impossible); participants with acute injuries, postoperative conditions or with serious comorbidities, such as Alzheimer's disease, Parkinson's disease, heart failure, liver/kidney disease, or diabetes mellitus; interventions that could not be classified as resistance training (e.g., Pilates, Yoga, and dance exercise).

Assessment of Methodological Quality of the Studies

The methodological quality (including risk of bias) of the included studies was independently rated by at least two researchers using the PEDro (Physiotherapy Evidence Database) scale [19]. The PEDro scale consists of 11 dichotomous (yes or no) items, in which criteria 2–9 rate randomization and internal validity and the criteria 10–11 rate the presence of statistically replicable results. Criterion 1 relates to the external validity and is not been used to calculate the PEDro score sum. Studies with a PEDro score ≥ 6 from 0 to 10 [19] represent a high-quality study.

Data Extraction

Relevant data were extracted independently by at least two researchers using a standardized extraction Excel spreadsheet adapted from the Cochrane Collaboration [20]. Means and standard deviations of pre- and posttest scores on relevant outcome parameters were extracted along with the number of participants assessed in each group. If these point and variability measures were not reported in the full-text article, the means and pooled within-group standard deviations of the change scores were entered in an electronic spreadsheet. If studies only presented means and standard deviations in figures, the WebPlotDigitizer Version 4 (Free Software Foundation, Boston, MA, USA) was used to extract means with

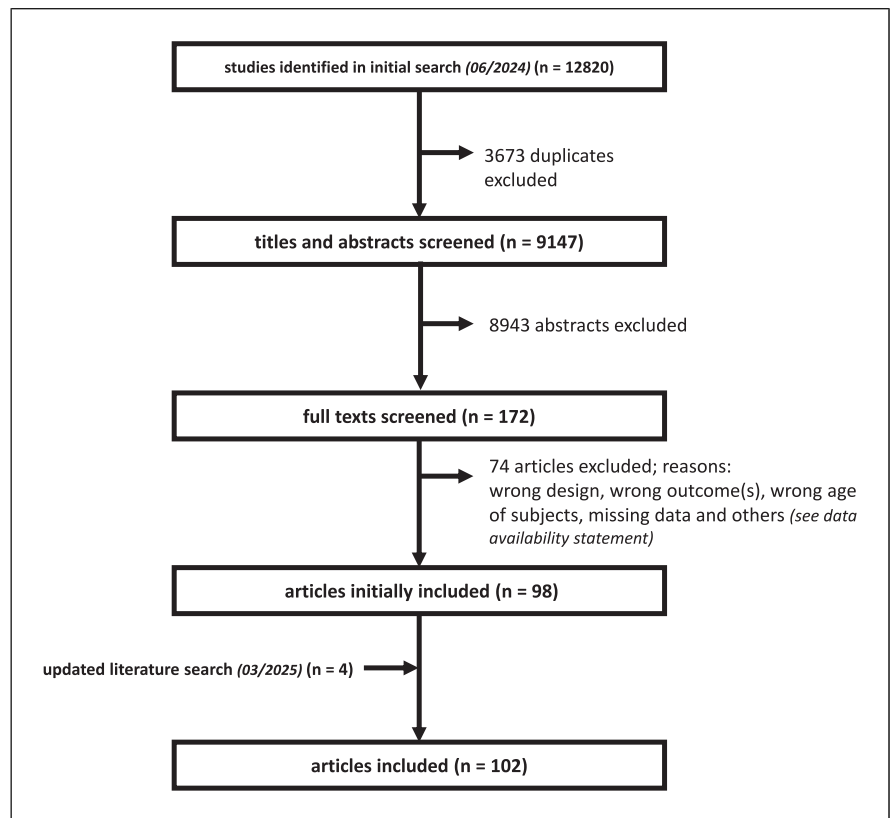


Fig. 1. Flowchart of the different phases of study screening and selection.

standard deviations from the figures [21]. If none of the methods provided sufficient data, the corresponding authors of eligible studies were contacted and missing values were requested twice within a 2 month period. Additionally, relevant study information regarding author(s), year of publication, number of participants, data on the training intervention (duration of the training period, training frequency, duration per session, training mode), control condition and PEDro scores were also recorded. For the purpose of the analysis, similar interventions were grouped into seven distinct interventions for comparison in the network: no intervention, bodyweight training, elastic band training, free weight training, machine-based training and mixed training. Thereby, an intervention was classified as bodyweight training when no external load apart from the bodyweight was used for a majority of relevant exercises. An intervention was categorized as elastic band, free weight or machine-based training when a majority of the relevant exercises were performed with elastic band resistance, free weights (barbells, dumbbells, or similar), or weight training machines, respectively. Finally, an intervention was classified as mixed training if more than one resistance training mode was used for a substantial part of the intervention.

Statistical Analysis

The NMA was conducted using a frequentist framework. This approach was chosen for its wide acceptance and familiarity in the field of exercise science. Frequentist methods facilitate the interpretation and communication of the findings for readers who may not be familiar with the complex nature of NMA. All calculations and presentational figures were computed using the R software (version 4.1.1; The R Foundation for Statistical Computing) and the package “netmeta” [22]. The standardized mean differences (SMDs) and 95% confidence intervals (CIs) were calculated for all interventional treatments as a measure of treatment effectiveness. SMDs were calculated as differences between means divided by the pooled standard deviations (trivial: $SMD < 0.2$, small: $0.2 \leq SMD < 0.5$, moderate: $0.5 \leq SMD < 0.8$, large $SMD \geq 0.8$) [23]. Subsequently, a network model for strength effects was computed. In order to visualize the networks, a network graph was created. The estimations of treatment effects were calculated based on a random-effects model in which “no intervention” was defined as the reference treatment [24]. A ranking was created based on the P-score of the individual treatments. The P-score represents the means of one sided p values under

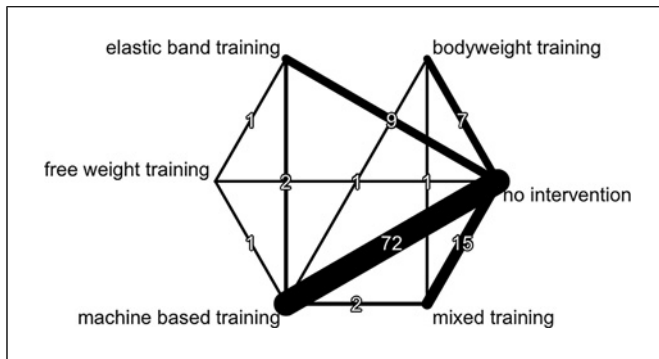


Fig. 2. Network plot with the different training modalities. The lines display a direct comparison. The width and number on the lines represent the number of direct comparisons.

normality assumption in a frequentist NMA [16]. It is interpreted as the mean extent of certainty that one intervention is superior to any other and is analogous to the surface under the cumulative ranking curve [17] values of a Bayesian NMA [16]. Additionally, a forest plot was created to further visualize the ranking and effects of the treatments. The decomposed Q statistics (within and between designs) were used to interpret potential heterogeneity and inconsistency. Heterogeneity was further quantified by I^2 [25]. A funnel plot was created to assess potential publication bias.

Results

In total, 102 included studies comprising 112 eligible pairwise comparisons enrolled 4,754 community-dwelling seniors (Fig. 1). The included studies had an average of 43 ± 32 (range: 12–275) participants with an average age of 70 ± 4 (range: 63–87) years. The mean duration of the included training interventions was 17 ± 12 (range: 4–72) weeks. The average study quality was rated high as indicated by a mean PEDro score of 6 ± 1 (range 3–8; online suppl. Table 2). The most common direct comparison of the network was no intervention vs. machine-based training ($n = 72$) [26–97] followed by no intervention vs. mixed training ($n = 15$) [30, 53, 94, 98–110], no intervention vs. elastic band training ($n = 9$) [61, 111–118], no intervention vs. bodyweight training ($n = 7$) [84, 109, 119–123], and multiple direct comparisons with either $n = 2$ or only $n = 1$ [53, 84, 109, 124–127] (see Fig. 2). Change scores and outcomes for the individual studies are depicted in online supplementary Tables 2 and 3 (S1, S2). The ranking of the training modalities is depicted in Figure 3 and effect sizes and 95%

CI are depicted in the forest plot in Figure 4. I^2 (75.3%; 95% CI: 70.0%, 79.6%) and Q statistics ($Q_{\text{within design}}: p \leq 0.0001$) revealed a considerable heterogeneity. Apart from this, Q statistics ($Q_{\text{between designs}}: p = 0.35$) revealed sufficient consistency across study designs. In addition, the funnel plot evaluation revealed a considerable risk of bias due to its asymmetry (Fig. 5).

Discussion

The aim of our approach was to expand upon current meta-analytic knowledge by conducting a NMA that examined both direct and indirect comparisons of different resistance training modalities (i.e., free weight training, machine-based training, bodyweight training, elastic band training, and mixed forms) and their effects on muscular strength outcomes in older adults. To the best of our knowledge, this is the first NMA comparing multiple different resistance training modalities in older populations. Consequently, it provides new insights into the comparative effectiveness of different forms of resistance training for older adults. Our findings based on 102 longitudinal training studies indicate that all of the examined resistance training modes elicit moderate to large effects on maximal strength gains in older adults when compared to a no intervention control. Additionally, all of the interventions seem to elicit rather robust effects, suggested by their 95% CIs. The modality ranking based on P-scores is in accordance with the observed effect magnitudes. Machine-based training elicits the largest effect, followed by free weight, elastic band, mixed, and bodyweight training. Bodyweight training stands out as the only intervention with a merely moderate effect in contrast to the large effects observed with the others.

Our NMA highlights the general effects of resistance training for strength gains in older adults and is therefore in line with common recommendations [8, 10] and other meta-analyses investigating resistance training for older individuals [6, 128]. For example, Grgic and colleagues [6] analyzed 22 studies to investigate the effects of resistance training on muscle strength in individuals over 75 years old. Similar to our NMA, they found a significant and large effect of resistance training in comparison with controls on muscle strength (effect size: 0.97; 95% CI: 0.50, 1.44), which was even larger in those aged 80 years and older. Similarly, another meta-analysis by Peterson and colleagues [128] analyzed 47 studies and also came to the conclusion that resistance training has a clear and robust effect on the strength of adults over the age of 50.

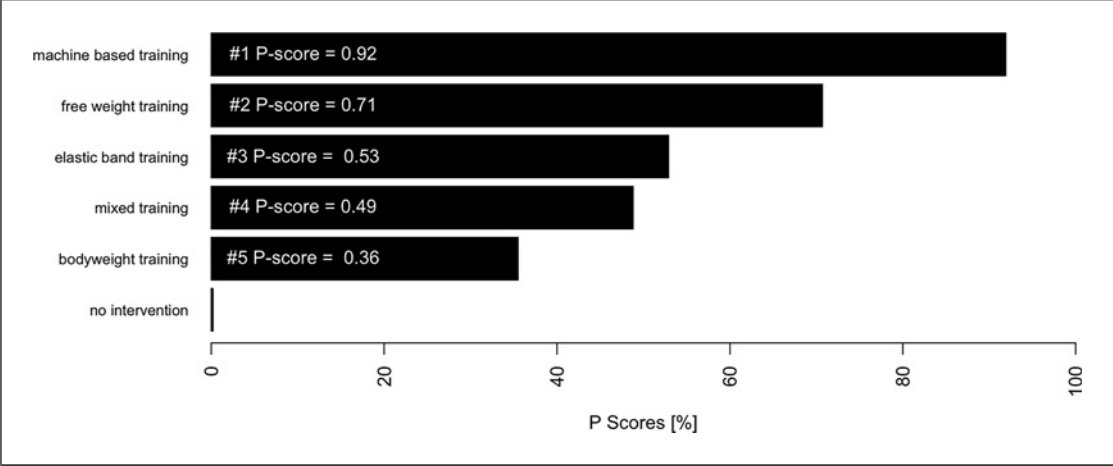


Fig. 3. P-score ranking of the different training modalities given as percentage. The P-score ranking as well as the P-score itself are displayed on the bars in white.

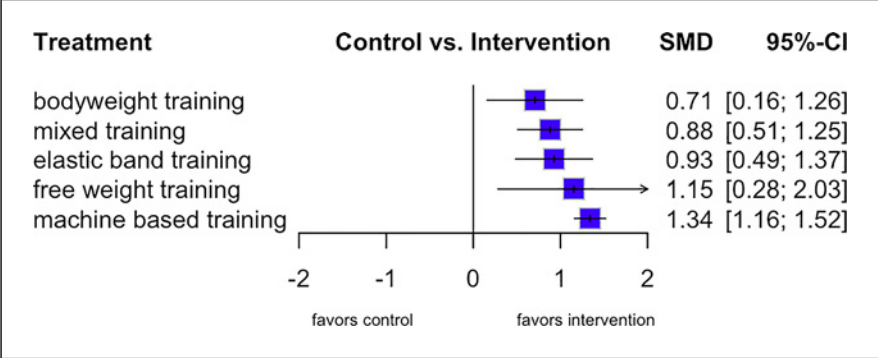


Fig. 4. Forest plot with the standard mean difference (SMD) and 95% confidence intervals (CIs) for maximal strength outcomes. Positive values indicate an effect in favor of the respective training modality, relative to no intervention.

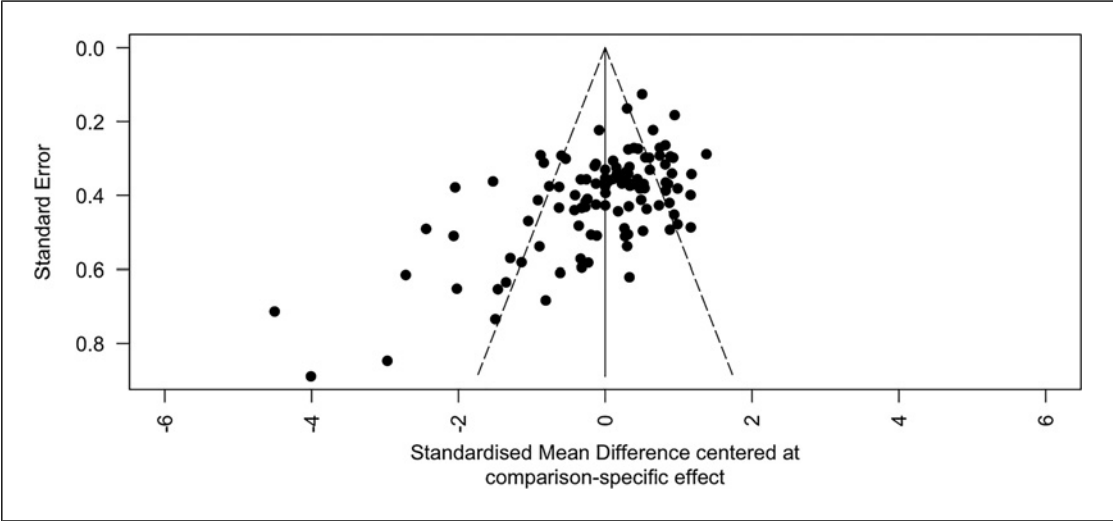


Fig. 5. Funnel plot for assessing publication bias. Each of the black dots represents an included study.

However, our network meta-analytical approach provides a perspective beyond previous findings by enabling the comparison of different training modalities. Our findings demonstrate substantial strength gains in older adults through resistance training regardless of the resistance type or equipment that was used. Differences in magnitude of the effects among the compared modalities suggest that although all interventions yielded moderate or large effects, machine-based training is considerably more likely to be effective than all of the other training modalities. This is further highlighted by the P-score ranking (Fig. 3) and can not only be attributed to the larger magnitude of the estimated effect but also to the noticeably smaller variance with machine-based training. Most likely, this is primarily due to the substantial difference in number of direct comparisons, which becomes particularly evident when comparing the two interventions with the largest effects (machine-based training: number of direct comparisons = 72; free weight training: number of direct comparisons = 3). This difference in sample size affects the variance and results in wider CIs for free weight training. It is unclear how the effects and P-scores of the two training modalities would compare if more studies, investigating free weight training, had been available for the analysis. Previous research comparing different resistance training modalities in a younger population up to 60 years are at least somewhat in agreement with the effect estimates demonstrated by our analysis [13, 129]. For example, a meta-analysis by Heidele and colleagues [13] from 2022 demonstrated that machine-based training and free weight training both resulted in non-different strength gains (ES: 0.128 [95% CI: -0.303, 0.559]). In another recent meta-analysis from 2023, Haugen and colleagues [129] displayed identical results. They found no difference between the two modalities for dynamic (SMD: 0.084, CI: -0.106, 0.273, $p = 0.387$) or isometric strength (SMD: -0.079, CI: -0.432, 0.273, $p = 0.660$) [129]. It is therefore reasonable to assume that free weight training would result in similar strength gains compared to machine-based training. Regarding the comparison between elastic band training and the rather conventional resistance training methods (free weight and/or machine training), a meta-analysis from Lopes and colleagues [11] found no superiority of either of the resistance training types (upper and lower limb: SMD = -0.011 and 0.09; $p = 0.48$ and 0.52). Our results are largely in line with earlier findings, because – like the other discussed modalities – elastic band training has a large effect on strength outcomes in older adults. In contrast to this, bodyweight training stands out as the only modality in our current NMA that

displays a merely moderate instead of a large effect. With seven included studies (number of direct comparisons = 9) that investigated bodyweight training [84, 109, 119–123], our results suggest slightly smaller effects of the training modality when it is compared to other resistance training modes. This is highlighted by the P-score ranking as well as the observed effect size. Other studies comparing the effects of bodyweight training with different resistance training modalities on strength gains vary in outcome [14, 15]. A study conducted by Martins and colleagues [15] from 2018 found less gains in leg extension 1RM strength for an interval based bodyweight resistance training compared to a combined resistance training with free weights and machines ($\Delta\% = 1.1$, SMD = 0.03 vs. $\Delta\% = 33.9$, SMD = 1.49). However, another study conducted by Kikuchi and Nakazato [14] demonstrated non-different effects on 1RM strength for a push up (SMD [95% CI] = 0.25 [-0.69; 1.17]) as well as a bench press training intervention (SMD [95% CI] = 0.41 [-0.54; 1.3]). Hence, although our results are not conclusive, a subtle trend can be assumed.

Nevertheless, in consideration of our findings, we suggest that there is a lack of compelling evidence to support a clear superiority of any single resistance training modality. Therefore, it seems worthwhile to investigate potential covariates that could account for the observed differences in effect magnitude. In this respect, it is well established that the training intensity (training load) of resistance training is a major factor influencing the effect magnitude of strength outcomes [130, 131]. This fact holds true for younger healthy trained and untrained populations and can likely be applied to older populations. In this regard, Csapo and Alegre [5] found that in older adults, high-intensity resistance training resulted in greater increases in muscle strength than low-to-moderate intensity training at a comparable training volume. Furthermore, Borde et al. [132] conducted a comprehensive review with meta-analysis on the effects of resistance training in older adults, including the investigation of dose-response relationships: Their results – based on 25 studies – indicated that resistance training showed a large effect (SMD = 1.57) on muscle strength. Moreover, their meta-regressions revealed that training intensity had significant effects on muscle strength. The highest effect sizes were found for the intensities of 70–79% of the 1RM (mean SMD = 1.89). With this in mind, it is of interest to compare training intensities used in the bodyweight training studies to those of the other interventions with larger effects. Unfortunately, load criteria are more frequently specified and standardized in interventions like machine-based

training or free weight training than in others like bodyweight or elastic band training. This is perhaps largely due to the inherent characteristics of the selected equipment. However, even with a lack of definitive quantifiable intensity specifications (e.g., the load relative to 1RM), most of the studies investigating bodyweight training reported efforts, loads or repetition schemes that allude to rather low training intensities (see online suppl. Table 2). In contrast, machine-based, free weight, or mixed resistance training frequently applied moderate intensities of 50–70% 1RM, high intensities of >70% 1RM or used a comparable effort and repetition scheme (e.g., repetition schemes of 5–13 reps accompanied by a rate of perceived exertion [RPE] ≥ 7 on the OMNI-Scale [133]), for substantial parts of the intervention (see online suppl. Table 2). Similarly, studies that investigated elastic band training also partially reported an effort (OMNI-Scale 7–10 out of 10) and repetition scheme (8–15 repetitions) potentially corresponding to a comparatively high training intensity (see online suppl. Table 2). Considering this, it is evident that all of the training modalities in our NMA that elicit a large effect largely include studies that incorporated moderate or high resistance training intensities for substantial parts of the training period. Besides intensity, further noteworthy load criteria that could potentially affect the magnitudes of strength gains elicited by the resistance training modalities are highlighted in the aforementioned meta-analysis by Borde and colleagues [132]. Their findings indicate that similar to training intensity and training volume, the duration of the training period had significant effects on development of muscle strength [132]. Adversely, our findings suggest study duration is not correlated with strength gain, when considered in isolation from other factors (online suppl. Fig. 1). Although a deeper analysis of load criteria is beyond the scope of this NMA, our findings, along with existing research, highlight the importance of thoughtful consideration of programming variables and load criteria when planning and implementing resistance training for older individuals. Assuming matching load criteria, such as training intensity, volume and time under tension, it appears likely that most training modalities could elicit similar strength gains in older adults over a given intervention duration [13, 129, 132]. However, certain resistance training modalities could still have distinct advantages in specific scenarios. In this respect, factors such as standardization, measurability of progressive overload, task specificity of the exercises and general accessibility should be taken into consideration [9, 10, 134]. Furthermore, research suggests that choosing the training modality by personal preference can present an

opportunity for optimizing adherence to an intervention [135, 136]. Studies indicate that increased enjoyment and motivation resulting from self-selection of exercises can lead to greater success of an intervention [135, 136]. Following this logic alongside the results of this analysis, it is likely that older adults would benefit from trying a variety of different resistance training types in order to determine their individual preferences and the best training modality for their specific scenario.

Two major limitations of this NMA are (1) the sparse availability of included studies for certain training modalities (e.g., free weight training) and (2) the lack of objective and comparable load criteria reported in certain studies that have been included. Especially for studies investigating resistance training in which the resistance is not constant or load cannot be quantified as easily (e.g., bodyweight training or elastic band training) the intensity had to be estimated from effort or similar parameters. Unfortunately, both of these limitations affect comparing and interpreting results across the different intervention types in our analysis, as they render a subgroup analysis unfeasible [137] and may also contribute to concealing the underlying cause of the observed heterogeneity in the meta-analysis. Furthermore, this lack of studies also prompted us to refrain from excluding studies solely based on low PEDro scores. In the future, additional high-quality RCTs with objective load criteria, such as a load relative to the 1RM or multiple RM, should be basis for a meta-regression or subgroup analysis which could help get a more nuanced perspective on the raised points in our discussion. Two additional minor limitations of our NMA are the substantial heterogeneity within study designs and the considerable risk of bias suggested by funnel plot asymmetry. Both of these limitations warrant careful interpretation of the results. However, given that our findings are rather conservative and robust, we believe that the practical risk of wrongful decision-making remains very low. This is why, despite these limitations, this NMA offers valuable insights by providing a comprehensive basis for decision-making in resistance training prescription for older adults and highlighting tendencies that are worth investigating in further meta-analyses.

In conclusion, our findings demonstrate that the effects of resistance training on strength gains are consistent across most of the investigated resistance training modalities. Machine-based training is well investigated and has the largest effect estimate out of all the training modalities. However, in comparison to no intervention, the analysis reveals large effects for all training modalities with the exception of bodyweight training, where the

observed effect is slightly smaller. Upon closer examination of the studies comprising the various modalities, there appears to be a trend, indicating that modalities with predominantly high training intensities elicit larger effects on strength compared to modalities in which training intensity is predominantly low. In consideration of our findings along with existing literature, we propose that, as long as load criteria such as training intensity, among others, are prescribed adequately, choosing the right training type might be a matter of scenario specific practicability, enjoyment, and individual preference.

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Statement of Ethics

An ethics statement is not applicable because this study is based exclusively on published literature.

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Conflict of Interest Statement

The authors report no conflicts of interest.

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Author Contributions

T.W., S.H., T.M., E.I., and L.D. conceptualized the study. T.W., E.B., and N.-H.H. with the help of others, screened the studies and extracted the relevant data. T.W., L.R., and S.H. performed the NMA. T.W. prepared the original draft. T.W., L.R., T.M., and L.D. reviewed and edited the draft. All authors read and approved the final manuscript.

Data Availability Statement

The data analyzed during the current study are available via the following public link: https://osf.io/jpe4f/?view_only=d2b40d0aa7e94b45bbd4b6338f9f2415.

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